AUTOMOBILE ENGINEERING · PART 3 OF 12

Transmissions and Drivetrain

By Isagi · 9 November 2026 · All specs as of November 2026 (manufacturer brochures and listings)

TL;DR

The gearbox exists because an engine works well only inside a narrow band of revs, while a car must move from a standstill to highway speed. Gears trade speed for pulling force. In India you will meet five kinds of transmission: manual, AMT, torque-converter automatic, CVT and dual-clutch (DCT), plus the e-CVT in strong hybrids. They differ in cost, smoothness and efficiency. The **drivetrain layout** (front-, rear- or four-wheel drive) decides which wheels get the power. Most Indian cars are front-wheel drive; true off-roaders need four-wheel drive with a low-range gear.

1. Why a car needs gears

A petrol engine delivers useful power roughly between 1,500 and 6,000 rpm. Idle is far lower, and a car's wheels turn at a few hundred rpm at city speeds. The transmission bridges the gap. **Torque at the driven axle = engine torque × gear ratio × final-drive ratio**, before losses. Take an engine making 100 Nm, a first gear of 3.45, a final drive of 4.3 and a fifth gear of 0.77 (illustrative figures, not a specific car): first gear delivers about 1,480 Nm to the axle and fifth about 330 Nm. First gear gives a strong push for starting and climbing; fifth lets the engine turn slowly at highway speed, which saves fuel and noise.

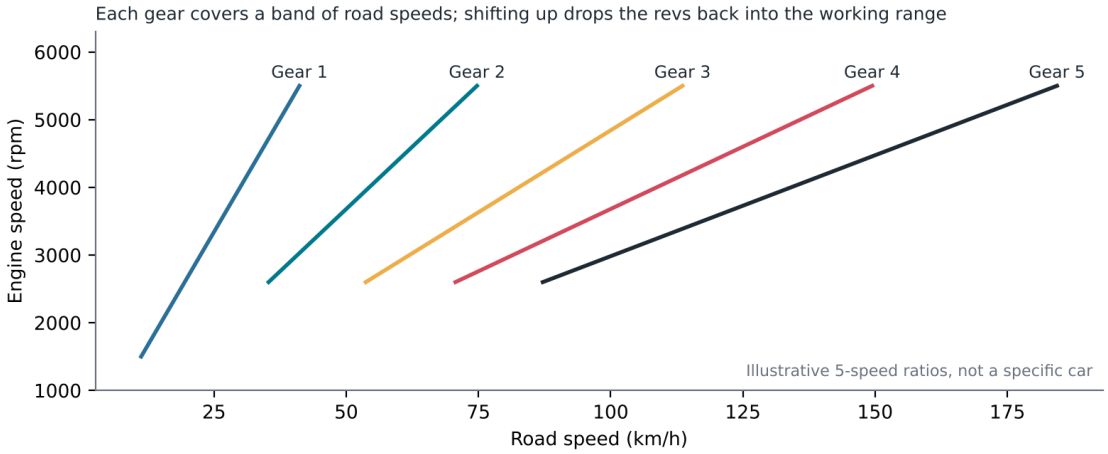


Fig 1. Each gear covers a band of road speeds. Shifting up drops the engine back to a lower rpm so it can keep accelerating.

More gears mean smaller steps between them, so the engine stays nearer its sweet spot. That is why 6-speed and 7-speed gearboxes help mileage compared with 4-speeds, and why the Part 2 point about wide torque plateaus matters: a wide plateau needs fewer shifts.

2. The manual: clutch and gears

A manual gearbox joins the engine to the wheels through a **clutch**, a pair of friction discs the driver presses apart or together. Pressing the pedal disconnects the engine so a gear can be chosen; releasing it reconnects. Manuals are cheap, light, efficient and durable, and they give the driver full control. Their weakness is the city: heavy traffic means constant clutch work. That is why automatic gearboxes now sell in large numbers in Indian metros.

3. The four kinds of automatic

All automatics take the clutch and gear selection away from the driver. They do it in four different ways, and the way decides how a car feels and what it costs to own.

Type	How it works	Strength	Weakness	Indian examples
AMT (automated manual)	A manual gearbox with a computer-controlled robot working the clutch and shifts	Cheapest automatic; manual-like mileage	Slow, jerky shifts, especially when pulling away	Maruti Swift AMT, Tata Punch AMT
Torque-converter automatic	A fluid coupling in place of the clutch, plus planetary gear sets	Smooth, tough, good for heavy loads and off-road	Heavier, costlier, a little less efficient	Creta diesel 6-speed automatic; Thar
CVT (continuously variable)	A belt running between two cone-shaped pulleys with no fixed gears	Very smooth; keeps revs in the best band	Rubbery feel when accelerating hard; less suited to high torque	Hyundai IVT (Creta petrol); Honda City
DCT (dual-clutch)	Two clutches, one for odd gears and one for even; the next gear is pre-selected	Fast shifts, sporty, efficient	Can be hesitant at crawling speed; costly to repair	Creta 1.5 turbo (7-speed DCT); Skoda 1.5 TSI (7-speed DSG)

The e-CVT in hybrids is a different idea: it uses a planetary gear set to blend power from the petrol engine and one or two electric motors, with no belt and no clutch. Toyota's Innova Hycross hybrid and the Hyryder use this layout, and it is why they feel smooth and need no shifts. Part 7 explains hybrids in full.

iMT (Hyundai and Kia's clutchless manual) is a halfway option: it has a manual gearbox with a sensor that handles the clutch, so you shift yourself but need no pedal.* Which type suits you depends on traffic, budget and how much you drive; Part 11 returns to that choice.

4. Drivetrain layouts: which wheels get the power

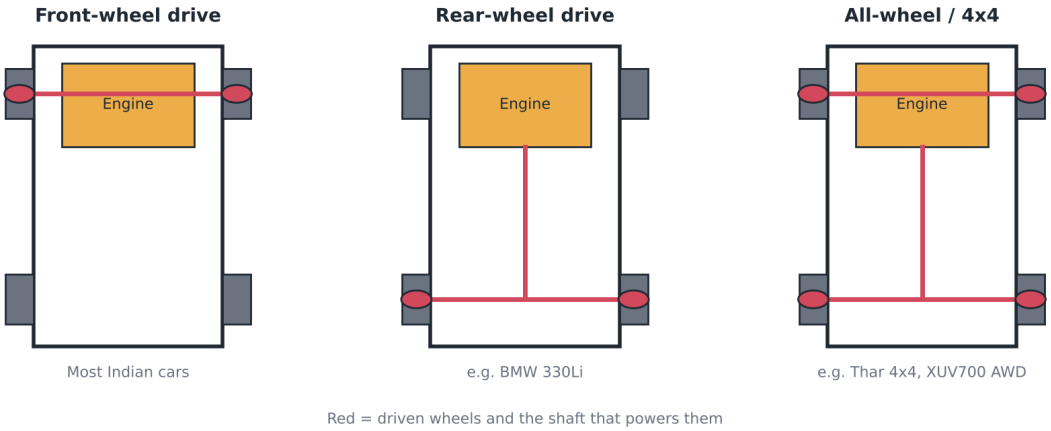


Fig 2. Three layouts. Red shows the driven wheels and the shaft that feeds them.

- **Front-wheel drive (FWD):** the engine and driven wheels sit together. It is cheap, compact and light, with good traction on wet roads. Almost every Swift, Nexon, Creta and City is FWD.
- **Rear-wheel drive (RWD):** a shaft carries power to the rear. It gives balanced weight and better handling at the limit, but it takes up floor space. The BMW 330Li is RWD; the Mahindra Thar offers RWD on some versions.*
- **All-wheel and four-wheel drive (AWD/4x4):** power goes to all four wheels. AWD systems (Mahindra XUV700 AWD*) work automatically on roads; a true 4x4 (Thar 4x4*) adds a *low-range* gearbox that multiplies torque again for steep, rough ground. Both add weight, cost and a fuel penalty.

A **differential** sits between the driven wheels so the outer wheel can turn faster than the inner one in a corner. An open differential sends power to whichever wheel slips first; *locking* differentials force both wheels to turn together, which is what off-roaders use on loose ground.

5. What it means for a buyer

- **City use, heavy traffic:** an automatic, preferably CVT, torque converter or DCT, is worth the premium; an AMT is the budget option.
- **Highway use:** a DCT or torque converter gives quick overtakes; a manual is cheaper and often a little more efficient.
- **Hills and rough roads:** torque-converter automatic or manual with strong low-end torque; for serious off-roading choose a 4x4 with low range.
- **Mileage:** manual ≥ AMT ≈ DCT ≥ CVT ≥ torque converter in most cases, but differences are small next to driving style (Part 2).*

Summary

- The transmission lets an engine with a narrow rev band move a car across all speeds; torque at the axle = engine torque × gear ratio × final drive.
- Manuals are cheap and efficient; automatics trade some of that for traffic comfort.
- AMT is a robot-driven manual: cheap but jerky. Torque converter is smooth and tough. CVT keeps revs steady. DCT is quick and sporty. e-CVT blends engine and motors in hybrids.
- FWD is common and cheap; RWD helps handling; AWD/4x4 helps traction and off-road ability, at a cost in weight and fuel.
- Choose by traffic, budget and terrain, not by the number of gears.

*Unverified or general knowledge: gear ratios in Section 1 are illustrative, not from a brochure; RWD/4x4 availability on the Thar and the XUV700 AWD, the iMT description and the mileage ordering in Section 5 are not checked against official documents. Transmission offerings confirmed in November 2026 listings: Swift 5-speed AMT; Creta petrol 6MT or IVT, turbo 7-speed DCT only, diesel 6MT or 6-speed automatic; Thar 6-speed torque-converter automatic; Honda City CVT; Hyryder e-CVT; Skoda 1.5 TSI 7-speed DSG only. Punch AMT and Innova Hycross e-CVT were not re-verified. Sources: Autocar India, Honda Cars India, Wikipedia. Theory: Heywood; Gillespie, Fundamentals of Vehicle Dynamics.

See also: the rest of the Automobile Engineering series (parts 1 to 12). Next: Part 4.